PORTFOLIO · SOFTWARE ENGINEERING | BACKEND & DATA

# Tiansawang Suetrong

Digital Science and Technology student at the Faculty of ICT, Mahidol University. I build full-stack systems end to end, from the interface people use to the API and the data model underneath. Two of them are live today.

INSIDE


01 **WasmBot Studio** No-code business rule bots on Rust WebAssembly

02 **HEAT//SHIFT** Map-first urban heat decision platform for Bangkok

03 **More work and contact** Portfolio site, practice projects, skills


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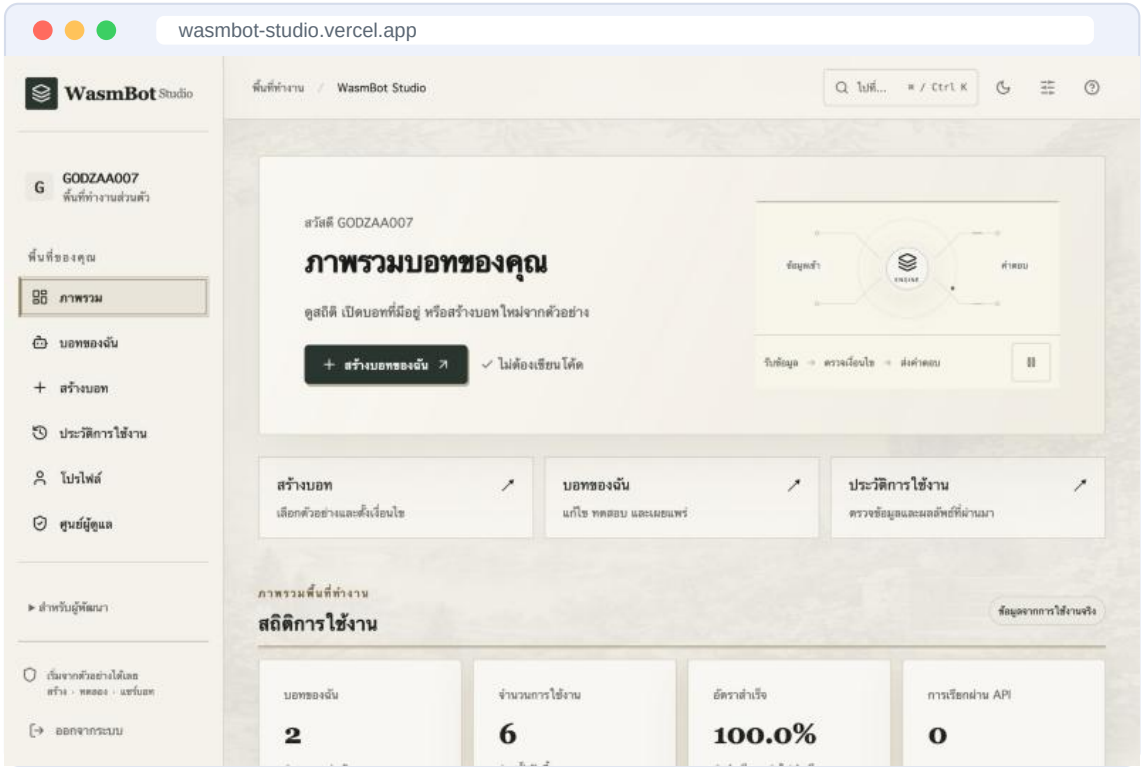


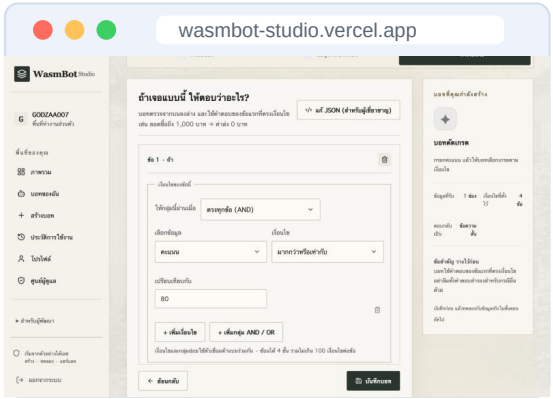

**Mahidol University · ICT**

B.Sc. Digital Science and Technology · Bangkok

01 / FULL STACK · NO-CODE PLATFORM

# WasmBot Studio




**Rule editor.** Nested AND/OR groups with 12 comparison operators. Both screenshots were captured from the live app on 11 November 2026.

A no-code platform that turns business rules into reusable bots: design them visually, test them on a real Rust WebAssembly engine, publish immutable versions, then use them anywhere.

## MY ROLE

Full-stack development: interface, Edge Functions, database, security, and deployment.

## WHAT I BUILT

- Visual AND/OR rule editor with **12 operators**, executed by a Rust engine compiled to WebAssembly.
- Immutable, versioned deployments so draft edits never affect the live bot.
- Each bot works as a hosted form, embed, REST API, signed webhook, and typed TypeScript SDK.
- Row-level security, scoped API keys stored as HMAC-SHA256 hashes, rate limits, and audit logs.
- **7 Supabase Edge Functions**, verified with **71 automated tests**.

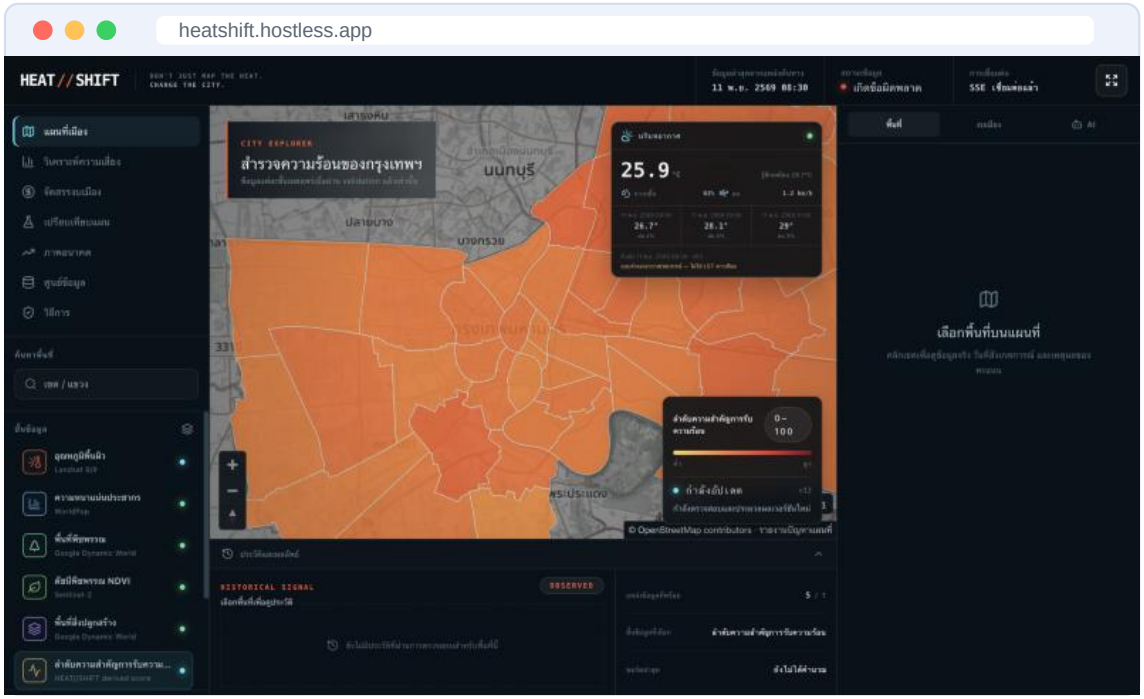
## STACK

TypeScript · React · Vite · Tailwind CSS · Supabase · PostgreSQL · Rust · WebAssembly

**Live app** wasmbot-studio.vercel.app (sign-in required)
**Source** github.com/tiansawang1818-pixel/wasmbot-studio

02 / GEOSPATIAL · DECISION INTELLIGENCE

# HEAT//SHIFT




## SEVEN REAL DATA SOURCES, VALIDATED BEFORE PUBLISHING

| | | | |
|---|---|---|---|
| **BMA boundaries**<br>50 Bangkok districts | **Landsat 8/9**<br>Surface temperature | **Sentinel-2**<br>Vegetation (NDVI) | **Dynamic World**<br>Land cover |
| **WorldPop**<br>Population | **Open-Meteo**<br>Hourly weather context | **OpenStreetMap**<br>Candidate public sites | **Traceable**<br>Every value versioned |

A map-first platform for planning where to invest in cooling Bangkok, built on real satellite, weather, population, and OpenStreetMap data across all 50 districts, with every number traceable to its dataset version.

### MY ROLE

System and data workflow design: pipeline, database, planning tools, and cloud deployment.

### WHAT I BUILT

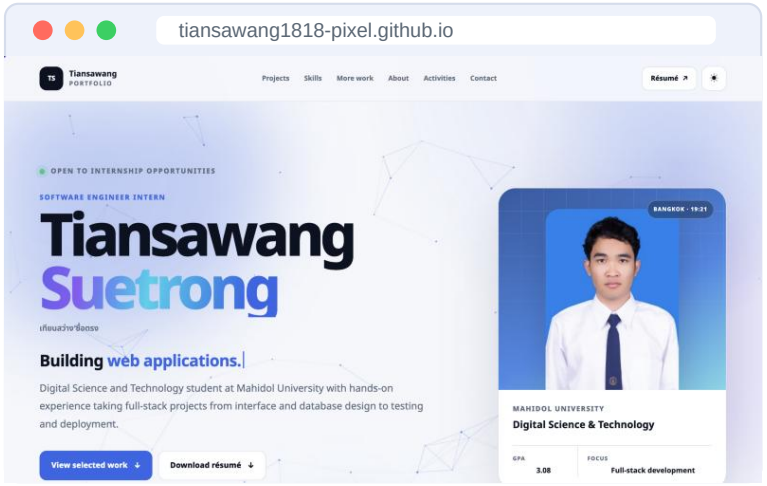
- **Six-stage pipeline**: staging, validation, processing, publishing, cache invalidation, live map updates.
- **Seven real sources** update automatically, including Landsat 8/9, Sentinel-2, Dynamic World, WorldPop, and OpenStreetMap.
- Versioned PostGIS datasets; invalid data never replaces the last good version.
- Budget optimizer with Google OR-Tools; forecasts that fail validation are withheld and their errors shown.
- Runs 24/7 on free-tier cloud hosting at **$0 cost**; admin tools behind single sign-on.

### STACK

Python · FastAPI · Next.js · MapLibre · PostgreSQL · PostGIS · Redis · Celery · Earth Engine

**Live app** heatshift.hostless.app
**Source** github.com/tiansawang1818-pixel/HEAT

# Smaller builds and practice

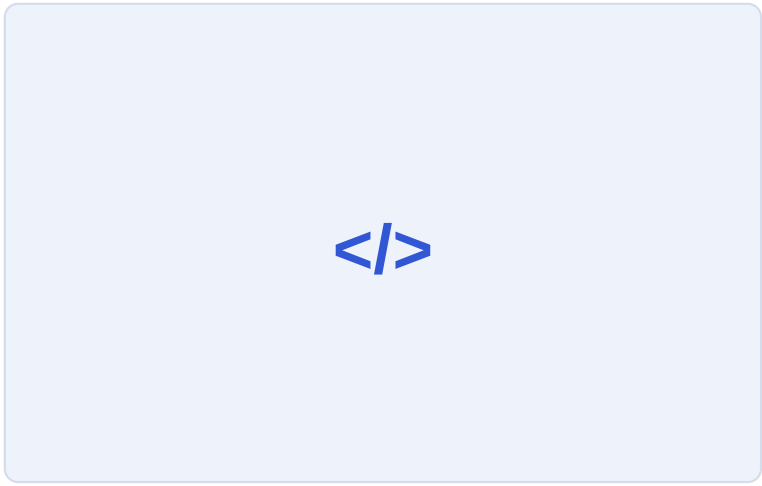



### This portfolio

HTML · CSS · JavaScript

Buildless, responsive site with dark mode, keyboard navigation, reduced-motion support, and a print layout.

View repository




### HTML101

HTML · CSS

Web fundamentals labs covering semantic HTML, page layout, styling, and multi-page sites.

View repository

## SKILLS

**Programming:** Python, TypeScript, JavaScript, Java, SQL
**Web Development:** HTML, CSS, Node.js, Web Services
**Databases:** PostgreSQL, MySQL, Supabase
**Data and Modeling:** ER modeling, relational database design, data validation
**Tools:** Git, GitHub, VS Code

## LET'S TALK

Looking for a software engineering, backend, or data internship. Email is the fastest way to reach me.

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